

Fundamentals of Software Project Management (SE4002)

Date: Sep 22nd 2025

Course Instructor(s)

Mr.Minhal Raza

Mid-1 Exam

Total Time(Hrs): 01

Total Marks: 28 Points

Total Questions: 03

Roll No

Section

Student Signature

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Attempt all the questions.

CLO #2: Critically evaluate and discuss the issues around project management and its application in the real world with course participants and learners

Q1 : Read and comprehend the following case study and answer the questions at the end of the case study
[10-Points]

Ali and Sara, two young entrepreneurs from Pakistan, have observed that many university students face difficulties in finding affordable and quality tutoring. Private tuition is often expensive, while free resources online are scattered, outdated, or not aligned with the local curriculum. To solve this problem, they plan to launch **EduLink**, a digital platform that connects students with qualified tutors online. The service will offer live tutoring sessions, recorded lectures, and smart study recommendations powered by AI. Their main target is university and college students who want flexible, affordable, and trustworthy academic support. EduLink will be available through a mobile app and website. To spread awareness, Ali and Sara plan to promote it on social media and also approach universities for partnerships. Parents who want reliable tutors for their children are also considered part of the audience. The platform will include features like a 24/7 chatbot for support, tutor ratings and reviews, and loyalty discounts for regular users. Students can pay through subscription plans, semester packages, or by booking single tutoring sessions. Tutors will also pay a small commission to EduLink for using the platform. Behind the scenes, Ali and Sara know they will need strong technology, a wide pool of tutors, and secure payment systems. They plan to collaborate with universities, telecom companies (for affordable internet bundles), and mobile payment services like Easypaisa and JazzCash. Of course, there are costs: developing and maintaining the platform, paying tutors, running marketing campaigns, and handling customer support. But Ali and Sara are confident that EduLink can become a go-to platform for affordable higher education support in Pakistan. You are hired as a software project manager and responsible for **creating business canvas model** for the EduLink. Be concise and clear.

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Answer :

Sustainable Business Model Canvas

Key Partners Universities (for collaborations) Tutors (qualified teaching pool) Telecom companies (cheap internet bundles) Mobile payment services (Easypaisa, JazzCash)	Key Activities Develop and maintain mobile app website Provide live tutoring recorded lectures Run AI-based smart study recommendations Marketing promotions on social media Manage payments and customer support Key Resources Strong technology (servers, app, website) Wide pool of qualified tutors Secure payment systems AI recommendation system Customer support team	Value Propositions Affordable and quality tutoring Flexible (live sessions + recorded lectures) Trustworthy (ratings reviews) 24/7 support chatbot Loyalty discounts	Customer Relationships Personalized tutor recommendations Ratings feedback system Customer support (chatbot + human help) Discounts loyalty rewards 5. Channels Mobile app Website Social media marketing University partnerships	Customer Segments University college students (main focus) Parents (for reliable tutors for children)
Cost Structure App website development/maintenance Tutor payments Marketing campaigns Customer support costs		Revenue Streams Subscription plans Semester packages One-time tutoring sessions Commission from tutors		

Q2 :Read and comprehend the following case study and answer the questions at the end of the case study **[10-Points]**

FAST University has decided to develop a mobile application called SmartLearn for its students. The app is expected to provide access to course materials such as PDFs, slides, and recorded lectures, along with a discussion forum where students and teachers can interact. It should also support online quizzes, grade tracking, and deliver notifications about deadlines, exams, and university events. The Dean of IT, acting as the project sponsor, has allocated a budget of PKR 8 million and set a timeline of six months for completion. A small team of developers, UI/UX designers, and QA testers has been assembled to execute the project. During the initial planning discussions, some challenges surfaced. Several faculty members requested additional features, such as live video streaming, which was not part of the original plan. Students also demanded integration with Google Drive and WhatsApp for easier file sharing. On the technical side, the IT team raised concerns about data security and the need for strong user authentication mechanisms. Despite these pressures, the Dean emphasized that the first version (Phase 1) must only include the core features outlined in the original plan. Advanced features like live video streaming may be considered for Phase 2, provided the budget allows.

- Identify in-scope and out-of-scope features for Phase 1.

In-Scope (Phase 1 – core features):

Access to course materials (PDFs, slides, recorded lectures)

Discussion forum (students + teachers)

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Online quizzes
Grade tracking
Notifications (deadlines, exams, events)
User authentication & data security

Out-of-Scope (Phase 1 – maybe in Phase 2):

Live video streaming
Google Drive integration
WhatsApp integration

ii. Write the best scope statement.

"The SmartLearn mobile application will provide FAST University students access to course materials (PDFs, slides, recorded lectures), online quizzes, grade tracking, and a discussion forum for student-teacher interaction. The app will also deliver notifications for deadlines, exams, and university events. Phase 1 will focus only on these core features within a budget of PKR 8 million and a six-month timeline. Additional features such as live video streaming and third-party integrations will be considered in future phases if resources allow."

iii. Create a scope baseline (WBS outline, deliverables, acceptance).

WBS Outline (Phase 1 only – high level):

SmartLearn App Development

- 1.1 Requirements gathering
- 1.2 UI/UX design
- 1.3 Core features implementation
 - 1.3.1 Course material access
 - 1.3.2 Discussion forum
 - 1.3.3 Online quizzes
 - 1.3.4 Grade tracking
 - 1.3.5 Notifications
 - 1.3.6 Authentication & security
- 1.4 Testing & QA
- 1.5 Deployment
- 1.6 Training & support

Deliverables

Mobile application (Android/iOS) with core features
Secure authentication system
Tested and working discussion forum, quiz, grades, and notification features
User guide / help document

Acceptance Criteria

Students can log in securely.
Students can view and download PDFs, slides, and recorded lectures.
Forum works for posting and replying.
Online quizzes are functional with grades visible.
Notifications are sent correctly.
App is stable, user-friendly, and tested before launch.

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iv. Discuss scope creep risks and control mechanisms.

Scope Creep Risks (extra work sneaking in):

Faculty asking for new features (e.g., live streaming).

Students demanding integrations (Google Drive, WhatsApp).

Developers adding “nice-to-have” features without approval.

Control Mechanisms:

Stick to scope baseline – only core features in Phase 1.

Use formal change request process – any new feature must be approved by Dean (sponsor).

Clear communication with faculty and students: extra features = Phase 2.

Regular scope reviews in project meetings.

CLO #1: Explain principles of the project life-cycle and how to identify opportunities to work with learners on relevant and appropriate project scenarios to share this understanding

Q3 : Read and comprehend the following case study and answer the questions at the end of the case study **[8-Points]**

1. The government has initiated a project to develop a Hospital Management System (HMS) for public hospitals. The project involves a large team of more than 150 members, including developers, testers, domain experts, and government officials. Since it is a long-term project lasting over three years, the system must comply with strict data privacy regulations such as HIPAA. One major challenge is that there are frequent change requests from different departments like doctors, nurses, and administrative staff, which makes requirement tracking complex. Moreover, there are several interdependent modules, such as patient records, pharmacy, and billing, that must be properly synchronized. The team is currently facing issues related to miscommunication, poor requirement tracking, and difficulty in managing dependencies.

2. A small startup is working on developing a mobile app with an AI chatbot that provides mental health support. The team is small, with only eight members, including developers, a psychologist, and a designer. Since the project has a tight budget and a 6-month deadline, cost-effective solutions are required. The development team prefers agile sprint planning, but the psychologist—who plays a critical role in designing the app’s conversational flow—finds complex task boards difficult to understand. The team is struggling to balance technical needs with non-technical usability, while also ensuring that they do not overspend on expensive enterprise project management tools.

For both cases separately , answer the following questions :

a. Which project management tools could be used here?

Case-1 : Jira → for requirement tracking & managing change requests.

Microsoft Project → for Gantt charts, scheduling, and dependencies.

Case-2 :

Trello → simple, visual boards (easy for non-technical people).

Asana → for small team task management.

Jira → (but it may be too complex for them).

Slack / Google Meet → for team communication.

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b. Which is the best choice and why?

Case-1 : Microsoft Project + Jira (together) are the best.

Microsoft Project helps manage long schedules, dependencies, and big teams. Jira helps track frequent requirement changes and issues. Together, they reduce miscommunication and keep requirements + timelines under control.

Case-2 : Trello is the best choice. It is free/low-cost (good for tight budget). Simple boards (easy for the psychologist to use). Supports agile sprints without being complicated. Helps balance technical + non-technical needs.

Decision Boundary / Short Summary :

Case 1: Hospital Management System (HMS) Big project with many people & rules → Microsoft Project + Jira.

Large team (150+ people)

Long project (3+ years)

Strict regulations (HIPAA, data privacy)

Frequent change requests + complex dependencies

Case 2: AI Chatbot Mobile App (Startup) Small startup with low budget → Trello (simple and cheap).

Small team (8 people)

Short project (6 months)

Low budget

Non-technical psychologist struggles with complex tools